# Iwasawa Main Conjecture and Kolyvagin's System of Gauss Sums

Aleksei Samoilenko

4 March 2026

# Overview

1. Class groups of cyclotomic fields
2. Gauss sums and Stickelberger relation
3. p-adic L-functions
4. Main Conjectures
5. Kolyvagin system of Gauss sums
6. Structure of A(F)
7. Proof of Main conjecture

# Class groups of cyclotomic fields

- Let $F = \mathbb{Q}(\zeta_p)$, $\Delta = \mathrm{Gal}(F/\mathbb{Q})$
- Consider $Cl(F)$ and it's $p$-primary part $A(F)$
- $\Delta$ acts on $Cl(F)$
- $Cl(F) = \bigoplus_{\chi} Cl(F)^{\chi}$
- $p$ is called irregular if $p \mid h$

- $h = h_F$ divisible into two factors

$$h^+ = \#Cl(F^+) = \prod_{\chi(-1)=1} \#Cl(F)^\chi$$

$$h^- = h/h^+ = \prod_{\chi(-1)=-1} \#Cl(F)^\chi$$

- Kummer-Vandiver conjecture predicts that $p \nmid h^+$

# Analytic class number formula

- Corollary from class number formula states

$$h^+ = (E^+ : C^+)$$

$$h^- = 2p \prod_{\chi(-1)=-1} B_{1,\chi}$$

- $E$-units of $F$, $C = E \cap \langle 1 - \zeta^k, \zeta^k \rangle$-cyclotomic units
- $E^+$-units of $F^+$, $C^+ = C \cap E^+$

# Notation

- Fix $M$ some big power of $p$
- Define $\mathbf{S}$=square free products of primes $\ell \equiv 1 \mod M$ (Always assume $n \in \mathbf{S}$)
- Fix $\delta \in \Delta$ and integers $b_n$ for each $n \in \mathbf{S}$ such that $\zeta_{nM}^{\widetilde{\delta}} = \zeta_{nM}^{b_n}$ for extension of $\delta$

# Gauss sums

- For $\mathfrak{r}$ above $r \equiv 1 \mod np$ Gauss sum $\mathfrak{g}(n, \mathfrak{r})$ for character $\varepsilon_{n,\mathfrak{r}}$ inverse to $n$-th power residue symbol modulo $\mathfrak{r}$

$$\mathfrak{g}(n, \mathfrak{r}) = \sum_{a=1}^{r-1} \varepsilon_{n,\mathfrak{r}}(a)\zeta_r^a \in F(\mu_{rn})$$

$$\alpha(n, \mathfrak{r}) = \mathfrak{g}(n, \mathfrak{r})^{\delta - b_n}$$

- $\alpha(n, \mathfrak{r}) \in F(\mu_n)$

# Stickelberger relation

- For $n \in \mathbf{S}$ define Stickelberger elements

$$\theta(n) = \frac{(\delta - b_n)}{np} \sum_{a=1}^{np} a\sigma_a^{-1} \in \mathbb{Z}[\mathrm{Gal}(F(\mu_n)/\mathbb{Q})]$$

**Theorem**

*We have decomposition of Gauss sum into primes, Stickelberger elements annihilate $\mathfrak{r}$ in class group*

$$\Big(\alpha(n, \mathfrak{r})\Big) = \mathfrak{r}^{\theta(n)}$$

# $p$-adic $L$-functions

- Consider values of Dirichlet $L$-functions at negative integers

$$L(1-n,\chi) = -\frac{B_{n,\chi}}{n}, \ \ n \geq 1$$

- $B_{n,\chi} = f^{n-1} \sum_{a=1}^{f} \chi(a) B_n(\frac{a}{f})$
- For $n \equiv n' \mod (p-1)p^n$ there is Kummer congruences

$$(1-p^{n-1})\frac{B_n}{n} \equiv (1-p^{n'-1})\frac{B_{n'}}{n'} \mod p^n$$

**Theorem**

*There exist unique p-adic meromorphic function $L_p(s, \chi)$ such that*

$$L_p(1-n, \chi) = (1 - \chi\omega^{-n}(p)p^{n-1})L(1-n, \chi\omega^{-n})$$

# Power series corresponding to $L_p(s,\chi)$

- Consider $\chi$-part of Stickelberger element: $\xi_m(\omega\chi^{-1}) = \frac{-1}{p^{m+1}} \sum_{a=1}^{p^{m+1}} a\chi(a)^{-1}\gamma_{m,a}^{-1}$
- $\theta_{p^{m+1}}(1)_\chi = -(\chi(b_1)\gamma_{m,b_1} - b_1)\xi_m(\omega\chi^{-1})$
- $\xi_m(\omega\chi^{-1}) \in \mathbb{Z}_p[\chi][\Gamma_m]$, and it's limit $\xi_\infty(\omega\chi^{-1}) \in \mathbb{Z}_p[\chi][[\Gamma]]$
- Because of isomorphism $\mathbb{Z}_p[\chi][[\Gamma]] \simeq \mathbb{Z}_p[\chi][[T]]$ there is a $G_p(\omega\chi^{-1}, T)$

**Theorem**

$L_p(s,\chi) = G_p(\chi, (1+p)^s - 1)$

# Characteristic ideal

- Define $X = \varinjlim A(F(\mu_{p^m}))$ under norm homomorphisms
- By Iwasawa classification of $\mathbb{Z}_p[[T]]$-modules $X^\chi \simeq \bigoplus_{i=1}^{n} \mathbb{Z}_p[[T]]/(f_i^\chi)$
- $char(X^\chi) = f^\chi = \prod_{i=1}^{n} f_i^\chi$

# Main Conjectures

**Theorem**

*For every odd character $\chi$ of $\Delta$*

$$char(X^{\chi}) = \left(G_p(\omega\chi^{-1}, T)\right)$$

**Theorem**

*For every even character $\chi$*

$$char\left(X^{\chi}\right) = char\left((\bar{E}_{\infty}/\bar{C}_{\infty})^{\chi}\right)$$

*where $\bar{E}_{\infty}$ and $\bar{C}_{\infty}$ are closures of limits of $E_m$ and $C_m$ respectively*

# $p = 37$

- It's known since Kummer that first irregular prime is $p = 37$, $h(\mathbb{Q}(\zeta_{37})) = 37$
- It occurs at $\chi = \omega^5$, $A^\chi = \mathbb{Z}/37\mathbb{Z}$
- Main conjecture predicts that there is a root of $L_p(s, \omega^{1-5}) = L_p(s, \omega^{32})$
- Corresponding unique root is
  $s = 13 + 20 \cdot 37 + 30 \cdot 37^2 + 8 \cdot 37^3 + 11 \cdot 37^4 + 12 \cdot 37^5 \cdots$

- Using Kolyvagin system of Gauss sums it's possible to give direct proof of the first form of Main Conjecture

# Euler system of Gauss sums

**Theorem**

*If $\ell$ is prime s.t. $\ell \mid n$, then*

$$\alpha(n, \mathfrak{r})^{\mathrm{N}_\ell} = \alpha(n/\ell, \mathrm{N}_\ell\, \mathfrak{r})^{1-Fr(\ell)^{-1}}$$

*in* $F(\mu_{n/\ell})^\times / (F(\mu_{n/\ell})^\times)^M$

# Derivatives

- Define $\mathrm{D}_l = \sum_{i=0}^{l-2} i\sigma_l^i \in \mathbb{Z}[\mathrm{Gal}(F(\mu_\ell))/F]$ and $\mathrm{D}_n = \prod_{\ell|n} \mathrm{D}_\ell$
- Define the set of derived Stickelberger elements $\{\delta(n) \in (\mathbb{Z}/M)[\mathrm{Gal}(F/\mathbb{Q})]\}$ as

$$\delta(n)\,\mathrm{N}_n = \mathrm{D}_n\,\theta(n)$$

- $\mathrm{N}_n$ is a norm element from $F(\mu_n)$ to $F$

# Kolyvagin system of derivatives of Gauss sums

- For $n \in \mathbf{S}$ and prime $\rho$ of $F$ above $r \equiv 1 \mod np$ there exist unique $\kappa(n,\rho) \equiv \alpha(n,\mathfrak{r})^{e_\chi D_n} \mod (F(\mu_n)^\times)^M$
- $\{\kappa(n,\rho)\}_{n\in\mathbf{S}}$ form Kolyvagin system

**Theorem**

- *If* $\ell \nmid nr$, *then* $[\kappa(n,\rho)]_l = 0$
- $[\kappa(n,\rho)]_r = \delta(n)_\chi \rho_\chi$

# Finite to singular map

- Let $\nu_\lambda : \{x \in F^\times | [x]_\ell = 0\} \to \mathbb{Z}/(\ell - 1)$ be discrete logarithm
- Define

$$\varphi_\ell : \{x \in F^\times / M | [x]_\ell = 0\} \to (\mathbb{Z}/M)[\Delta]$$

$$\varphi_\ell(x) = \sum \nu_{\lambda^\sigma}(x)\sigma$$

- Consider $(\mathbb{Z}/M)[\Delta]$ as singular part

## Theorem

*$n \in \mathbf{S}$, $\rho$ and $\rho'$ primes in $F$ above distinct $r \equiv r' \equiv 1 \mod nM$ and there are primes of $\mathfrak{r}$ and $\mathfrak{r}'$ whose projections in class group of $F(\mu_n)$ are the same. Then*

$$\varphi_\rho(\kappa(n, \rho')) \equiv \delta(nr)_\chi \mod \delta(n)(\mathbb{Z}/M)[\Delta]_\chi$$

# Rigidity property

**Theorem**

*Let $k \in \mathbb{N}$ prime to $n$, $c \in A(F(\mu_n))_\chi$, finite $G$-submodule $W \subset (F^\times/M)_\chi$ and $G$-homomorphism $\psi : W \to \mathbb{Z}/M[\Delta]$. Then there are infinitely many primes $\lambda$ of $F$ s.t.*

- *There is a prime above $\lambda$ whose projection is $c$*
- $\ell \equiv 1 \mod nkM$
- $\forall w \in W$ $[w]_\ell = 0$, *and* $\varphi_\lambda(w) = u\psi(w)$ *for some* $u \in (\mathbb{Z}/M)^\times$

- Let $d(n) = gcd(M, \delta(n))$ - depth of $\delta(n) \in \mathbb{Z}/M[\Delta]_\chi$
- $d(1)(\mathbb{Z}/M) = B_{1,\chi^{-1}}(\mathbb{Z}/M)$

**Theorem**

*Let $c \in A^\chi$, $B \subset A$-subgroup generated by primes dividing $n$. If $d(n) \mid d(1)$, $c \notin B^\chi$, then there is a prime $\lambda$ of $F$ s.t.*

- *Projection of $\lambda$ is $c$*
- $d(nl) \mid d(n)$
- $\frac{d(n)}{d(nl)}$ *annihilates $c$ in $A^\chi/B^\chi$*

## Corollary

$$\#(A^\chi/B^\chi) \mid d(n)$$

## Theorem

*For every odd* $\chi \neq \omega$

$$\#(A^\chi) = p^{\mathrm{ord}_p(B_{1,\chi^{-1}})}$$

# Proof of Main conjecture

We will generalize what we done for $F$ to each level $F_m = F(\mu_{p^{m+1}})$

- From Iwasawa classification we have

$$0 \to \bigoplus_{i=1}^{k} \Lambda^{\chi}/f_i \to X^{\chi} \to Z \to 0$$

- Fix $\eta \in \Lambda^{\chi}$ s.t. $\eta Z = 0$ and prime to $(1+T)^{p^m} - 1$, $G_p(\omega\chi^{-1}, T)$
- For each $n \in \mathbf{S}$ define $T_n^{\chi}$ as ideal generated by $\{\delta(n') : n' \mid n\}$ in $\mathbb{Z}/M[\mathrm{Gal}(F_m/\mathbb{Q})] =: R_{m,\chi}$

# Core vertices

- Put $h_0 = \delta(1) \in T_1^{\chi}$

**Theorem**

*For each $j$ s.t. $1 \leq j \leq k$ there exist rational prime $r_j \equiv 1 \mod n_j M p^{2m}$ and $h_j \in T_{n_j}^{\chi}$ with $n_j = r_1 \cdots r_j$ s.t. $f_j h_j = \eta h_{j-1}$ in $R_{m,\chi}$*

## Theorem

*For each* $1 \leq j \leq n$ *there exist* $\vartheta_j^\chi$ *such that* $f_j^\chi = \vartheta_{j-1}^\chi / \vartheta_j^\chi$ *with* $\vartheta_0^\chi = G_p(\omega\chi^{-1}, T)$

- Using this theorem we have

$$f^\chi = \prod_j f_j^\chi = \vartheta_0^\chi / \vartheta_k^\chi = G_p(\omega\chi^{-1}, T)/\vartheta_k^\chi$$

- Hence for all $\chi$ we have $f^\chi \mid G_p(\omega\chi^{-1}, T)$
- But if we take product over all $T_n = \zeta_{p_n} - 1$ on the l.h.s. we obtain $h_{p^n}^-$ which is generate the same ideal in $\mathbb{Z}_p$ by the crollary of class number formula

# References


Aoki, M. (2001).
The iwasawa main conjecture and gauss sums.
*Journal of Number Theory.*

Forras, B. (2020).
Iwasawa theory.

Maurer, Y. (2000).
Zeros of p-adic l functions.

Rubin, K. (1991).
Kolyvagin's system of gauss sums.
*Progress in Mathematics*, 89(3):309–324.